

ECO310 Midterm Cheat Sheet

1 Preferences & Utility

Rational: Complete + Transitive

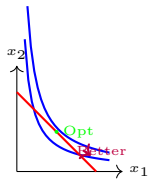
Monotone: More is better

Convex: Mixtures $\geq$ extremes (ICs bow inward)

1.1 MRS

$$MRS = \frac{MU_1}{MU_2} = - \frac{dx_2}{dx_1} \Big|_{IC}$$

At optimum: $MRS = p_1/p_2$



1.2 Budget Constraint

$$p_1 x_1 + p_2 x_2 = m$$

Slope: $-p_1/p_2$, Intercepts: $(m/p_1, 0)$, $(0, m/p_2)$

2 Utility Functions

2.1 Cobb-Douglas

$$u = x_1^a x_2^b$$

! Magic Rule: Spend $\frac{a}{a+b}$ on good 1, $\frac{b}{a+b}$ on good 2

$$x_1^* = \frac{a}{a+b} \cdot \frac{m}{p_1}, \quad x_2^* = \frac{b}{a+b} \cdot \frac{m}{p_2}$$

$$MRS = \frac{a}{b} \cdot \frac{x_2}{x_1}$$

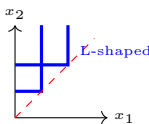
2.2 Perfect Complements

$$u = \min\{ax_1, bx_2\}$$

Always at kink: $ax_1 = bx_2$

$$x_1^* = \frac{bm}{bp_1 + ap_2}, \quad x_2^* = \frac{am}{bp_1 + ap_2}$$

! No substitution effect! (only income effect)



2.3 Perfect Substitutes

$$u = ax_1 + bx_2$$

$MRS = -a/b$ (constant)

! Buy all of cheaper good:

- If $\frac{a}{p_1} > \frac{b}{p_2}$: $x_1 = m/p_1$, $x_2 = 0$
- If $\frac{a}{p_1} < \frac{b}{p_2}$: $x_1 = 0$, $x_2 = m/p_2$

2.4 Quasilinear

$$u = v(x_1) + x_2$$

$MRS = v'(x_1)$ (only depends on x_1)

! No income effect for x_1 (at interior)

2.5 CES

$$u = (x_1^r + x_2^r)^{1/r}$$

$r \rightarrow 0$: CD; $r \rightarrow -\infty$: PC; $r = 1$: PS

3 Optimal Choice

3.1 Interior Solution

Tangency: $\frac{MU_1}{p_1} = \frac{MU_2}{p_2}$ or $MRS = \frac{p_1}{p_2}$

Budget: $p_1 x_1 + p_2 x_2 = m$

Solve both equations for x_1^*, x_2^*

3.2 Lagrangian

$$\mathcal{L} = u(x_1, x_2) + \lambda[m - p_1 x_1 - p_2 x_2]$$

FOCs: $MU_i = \lambda p_i$ for each i

λ = marginal utility of income

3.3 Corner Solution

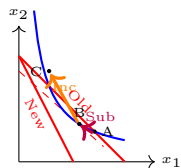
Check: $x_i \geq 0$ always!

If $MRS > p_1/p_2$ everywhere: buy only x_1

If $MRS < p_1/p_2$ everywhere: buy only x_2

4 Income & Sub Effects

Total = Substitution + Income



Sub effect: Always ≤ 0 (buy less when p rises) - Move along IC to new price ratio - = compensated bundle - original

Income effect: Sign depends on normal/inferior - Normal: < 0 (buy less) - Inferior: > 0 (buy more) - = new bundle - compensated

! Giffen: Income effect dominates, demand increases when price rises

5 Uncertainty

5.1 Expected Utility

$$EU(L) = \sum p_i u(c_i)$$

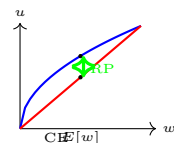
vNM utility: Unique up to $\bar{u} = a + bu$ ($b > 0$)

! NOT arbitrary transforms!

5.2 Risk Aversion

$$u'' < 0 \Leftrightarrow \text{Risk Averse}$$

Prefer certainty to gambles: $u(E[x]) > E[u(x)]$



CE: $u(CE) = EU$

RP: $RP = E[w] - CE$ (how much pay to avoid risk)

Risk averse: $RP > 0$

5.3 Arrow-Pratt

$$r_A(x) = - \frac{u''(x)}{u'(x)}, \quad r_R(x) = x \cdot r_A(x)$$

Higher $r_A \Rightarrow$ more risk averse

CRRA: $u = \frac{1-\rho}{1-\rho}$ or $\ln x$ - $r_A = \rho/x$ (DARA), $r_R = \rho$

CARA: $u = -e^{-\alpha x}$ - $r_A = \alpha$, $r_R = \alpha x$ (IRRA)

5.4 Insurance

Wealth w , loss D prob π , premium q , coverage K

$$EU = (1 - \pi)u(w - qK) + \pi u(w - D + K(1 - q))$$

Fair: $q = \pi \Rightarrow$ full insurance ($K = D$)

Unfair: $q > \pi \Rightarrow$ partial insurance ($K < D$)

FOC: $(1 - \pi)u'(c_1)(-q) + \pi u'(c_2)(1 - q) = 0$

5.5 Portfolio

Invest a in risky (return $\bar{r}$), rest at r_f :

$$\bar{W} = w(1 + r_f) + a(\bar{r} - r_f)$$

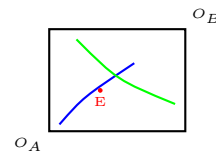
FOC: $E[u'(\bar{W})(\bar{r} - r_f)] = 0$

If $E[\bar{r}] > r_f$: invest $a^* > 0$

6 General Equilibrium

6.1 Edgeworth Box

2 consumers, 2 goods, total (E_1, E_2)



Feasible: $x^A + x^B = E$

PE: $MRS^A = MRS^B$ (ICs tangent)

6.2 Competitive Equilibrium

Each: $\max u$ s.t. $p \cdot x = p \cdot e$ (endowment)

Markets clear: $\sum x_i = E_i$

Walras' Law: If $n - 1$ markets clear, n -th clears

How to solve:

1. Set $p_2 = 1$ (numeraire)
2. Income: $m^i = p_1 e_1^i + e_2^i$
3. Demands: $x_1^i(p_1)$ from each UMP
4. Market clear: $\sum x_1^i(p_1) = E_1$
5. Solve for p_1^*

6.3 Welfare Theorems

1st: CE $\Rightarrow$ PE (needs LNS)

2nd: Any PE can be CE with transfers (needs convexity)

7 Duality

7.1 Expenditure Function

$$e(p, \bar{u}) = \min p \cdot x \text{ s.t. } u(x) \geq \bar{u}$$

Gives Hicksian demand: $h_i(p, \bar{u})$

Shepard's Lemma: $h_i = \frac{\partial e}{\partial p_i}$

7.2 Relations

$$x_i(p, m) = h_i(p, v(p, m))$$

$$h_i(p, \bar{u}) = x_i(p, e(p, \bar{u}))$$

$$v(p, e(p, \bar{u})) = \bar{u}$$

$$e(p, v(p, m)) = m$$

8 WARP

If x chosen when y affordable, and y chosen later, then y must NOT have been affordable when x chosen.

Check: For (p^A, m^A) and (p^B, m^B) :

- Is $p^B \cdot x^A \leq m^B$?
- Is $p^A \cdot x^B \leq m^A$?
- If both strictly: WARP violation

$\alpha + \beta \Leftrightarrow$ Rationalizable $\Leftrightarrow$ WARP

9 Quick Tips

! Always check $x_i \geq 0$!

! CD: Use fraction rule, don't solve manually

! PC: Zero sub effect, all income

! PS: Corner solution (all or nothing)

! QL: No income effect for x_1

! vNM: Only affine transforms allowed

! Sub effect: Always negative for own price

! Giffen: Rare - income dominates sub

! Risk averse: Concave u , $u'' < 0$

! Fair insurance: Full coverage optimal

! PE: $MRS^A = MRS^B$ at tangency

! Walras: Only need $n - 1$ market clearing

10 Problem Templates

10.1 UMP with CD

Given: $u = x_1^a x_2^b, (p_1, p_2, m)$

Solution:

$$x_1^* = \frac{a}{a+b} \cdot \frac{m}{p_1}, \quad x_2^* = \frac{b}{a+b} \cdot \frac{m}{p_2}$$

Done! (No calculus needed)

10.2 UMP General

- 1. $MRS = MU_1 / MU_2$
- 2. Set $MRS = p_1 / p_2$
- 3. Solve for $x_2(x_1)$ or $x_1(x_2)$
- 4. Plug into $p_1 x_1 + p_2 x_2 = m$
- 5. Solve for x_1^*, x_2^*
- 6. Check: both ≥ 0 ?

10.3 Income/Sub Effects

When p_1 changes from p_1 to p_1' :

- 1. **Original:** $A = (x_1, x_2)$ at (p_1, p_2, m)
- 2. **Utility:** $\bar{u} = u(A)$
- 3. **Compensated:** Solve $\min p_1' x_1 + p_2 x_2$ s.t. $u(x) = \bar{u}$
Get $B = (x_1^C, x_2^C)$
- 4. **New:** $C = (x_1', x_2')$ at (p_1', p_2, m)
- 5. Sub = $B - A$, Income = $C - B$

10.4 Expected Utility

Given: States $s = 1, \dots, n$ with probs p_s , outcomes c_s , utility u

- 1. $EU = \sum_s p_s u(c_s)$
- 2. If choosing x : write $c_s(x)$, $EU(x) = \sum_s p_s u(c_s(x))$
- 3. FOC: $\frac{dEU}{dx} = \sum_s p_s u'(c_s) \frac{dc_s}{dx} = 0$
- 4. Solve for x^*

10.5 CE and RP

Given: Lottery with EU , utility u

- 1. Find $E[w] = \sum p_i w_i$
- 2. Find $EU = \sum p_i u(w_i)$
- 3. Solve $u(CE) = EU$ for CE
- 4. $RP = E[w] - CE$

10.6 General Equilibrium

Given: 2 consumers with utilities u^A, u^B , endowments e^A, e^B

- 1. Normalize $p_2 = 1$
- 2. Incomes: $m^A = p_1 e_1^A + e_2^A, m^B = p_1 e_1^B + e_2^B$
- 3. Solve UMPs: $x_1^A(p_1), x_1^B(p_1)$
- 4. Market clear: $x_1^A(p_1) + x_1^B(p_1) = e_1^A + e_1^B$
- 5. Solve for p_1^*
- 6. Find allocations: $(x_1^A(p_1^*), x_2^A(p_1^*)),$ etc.
- 7. Check: all ≥ 0

11 Common Mistakes

- ! Mixing up $x(p, m)$ vs $h(p, \bar{u})$
- ! Sub effect: direction! (Less of good when price rises)
- ! PC: no sub effect (everything is income)
- ! vNM: can't use u^2 or $\sqrt{u}$ (only $a + bu$)
- ! Variance vs std dev (σ^2 vs σ)
- ! Corner: set $x = 0$, don't use FOC
- ! "Spend \$X" $\neq$ "Buy X units" (unless $p = 1$)
- ! Quasilinear: check $x_2 \geq 0$
- ! CE allocations: verify market clearing
- ! WARP: "revealed preferred" = chosen when other affordable

12 Key Formulas

12.1 Derivatives

$$\begin{aligned} \frac{d}{dx}(x^n) &= nx^{n-1} \\ \frac{d}{dx}(\ln x) &= \frac{1}{x} \\ \frac{d}{dx}(e^x) &= e^x \\ \frac{d}{dx}(e^{ax}) &= ae^{ax} \end{aligned}$$

12.2 Algebra

$$\begin{aligned} x^a x^b &= x^{a+b} \\ (x^a)^b &= x^{ab} \\ \ln(xy) &= \ln x + \ln y \\ \ln(x^a) &= a \ln x \end{aligned}$$

$$e^{\ln x} = x, \ln(e^x) = x$$

12.3 Optimization

FOC: $\frac{df}{dx} = 0$

SOC: $\frac{d^2 f}{dx^2} < 0$ (max), > 0 (min)

Constrained: Use Lagrangian

$$\mathcal{L} = f(x) + \lambda[g(x) - c]$$

FOC: $\frac{\partial \mathcal{L}}{\partial x} = 0, \frac{\partial \mathcal{L}}{\partial \lambda} = 0$

12.4 Probability

$$\begin{aligned} E[X] &= \sum_i p_i x_i \\ \text{Var}(X) &= E[X^2] - (E[X])^2 \\ \sigma &= \sqrt{\text{Var}(X)} \end{aligned}$$

13 Exam Strategy

- 1. **Read carefully:** What type of utility?
- 2. **CD shortcut:** Don't waste time on calculus
- 3. **Check corners:** Always verify $x \geq 0$
- 4. **Draw diagrams:** ICs + budget for intuition
- 5. **Sub effect:** Always negative for own price
- 6. **State-contingent:** Label states clearly
- 7. **Units:** Match dimensions (dollars, utils, etc.)
- 8. **Final check:** Does answer make economic sense?